

Startup Financial KPI Analysis

Introduction

This project focuses on analyzing key financial performance indicators (KPIs) of a startup using Python and Google Sheets. The objective is to evaluate business performance by examining critical metrics such as revenue, Customer Acquisition Cost (CAC), Lifetime Value (LTV), Burn Rate, and Run Rate. These KPIs help in understanding the financial health and sustainability of a startup. By leveraging data-driven techniques, the project provides insights into growth trends, cost efficiency, and overall business performance.

Abstract

The project involves processing startup data to compute important financial KPIs and visualize them using Google Sheets. Python is used for data cleaning, transformation, and calculation of metrics, while Google Sheets is used for dashboard creation and visualization. The analysis highlights revenue patterns, customer acquisition efficiency, and operational costs. By comparing LTV and CAC, the project evaluates whether the business model is sustainable. Overall, this project demonstrates how data analytics can support strategic decision-making in startups.

Tools Used

- **Python (Pandas, NumPy):** For data processing and KPI calculations
 - **Google Sheets:** For visualization and dashboard creation
 - **VS Code:** Development environment for coding and execution
 - **gsread & google-auth:** For integration between Python and Google Sheets
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Steps Involved in Building the Project

1. **Data Acquisition:**
Collected the startup dataset (`50_Startups.csv`) and prepared it for analysis.
 2. **Data Preparation:**
Cleaned and structured the dataset by renaming columns, handling inconsistencies, and ensuring data quality.
 3. **KPI Computation:**
Calculated key financial metrics including Customer Acquisition Cost (CAC), Lifetime Value (LTV), Burn Rate, Run Rate, and LTV:CAC ratio.
 4. **Cohort Structuring:**
Organized the dataset into time-based cohorts to analyze customer behavior and revenue trends.
 5. **Data Export:**
Generated a processed dataset and saved it as a CSV file for further use.
 6. **Integration with Google Sheets:**
Automated data upload using APIs to enable real-time access and collaboration.
 7. **Dashboard Development:**
Created visualizations such as revenue trends, burn rate charts, and LTV vs CAC analysis.
 8. **Insight Generation:**
Analyzed KPI trends to evaluate business performance and identify areas of improvement.
 9. **Validation and Quality Check:**
Verified accuracy of calculations and ensured consistency across all outputs.
 10. **Reporting and Documentation:**
Compiled the findings into a structured report suitable for academic and professional use.
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Conclusion

The project successfully demonstrates how financial KPI analysis can be used to evaluate startup performance. The results indicate that while the startup shows potential for growth, managing expenses and optimizing customer acquisition strategies are crucial for long-term success. The LTV:CAC ratio serves as a key indicator of business sustainability. By integrating Python with Google Sheets, the project provides a simple yet effective solution for financial analysis. This approach can be extended further by incorporating advanced analytics and predictive models.